

#1 PRIMARY GN → ESKD US

- 1/3 ADULT NEPHROTICS
- AFRICAN ANCESTRY 50%

PRIMARY PATHWAY

- NOS (Most Common)
- TIP LESION (Best Prognosis)
- CELLULAR HEAVY PROTEINURIA
- PERIHILAR OFTEN SECONDARY
- COLLAPSING WORST MY APOL I

RAAS + SGLT2i — ALL TYPES

APOLI > 4 MONTHS STEROIDS 1ST LINE

NEPHROTOXIC CNI

BEFORE STEROID-RESISTANT CAN ANCESTRY COLLAPSING

APOLI

PRIMARY + NEPHROTIC

GENETIC / SECONDARY PATHWAYS

- SECONDARY GENETIC: TREAT UNDERLYING**
- STOP** (No Steroids No IS)
- TEST IF:** FAMILY HX STEROID-RESISTANT
NPHS1, NPHS2, INF2, ACTN4, TRPC6, WT1
- PLEX** (RECURRENT P POST-Tx)
- TRANSPLANT GATE**: PRIMARY (\$\$K NECESS), SECONDARY fare ✓, GENETIC fare ✓
- ~50% ESKD in 10 YRS**
- SPONTANEOUS REMISSION RARE**

GO DEEPER

1. #1 cause adult ESKD nephrotic

WHAT YOU SEE

Banner '#1 PRIMARY GN → ESKD US / 1/3 ADULT NEPHROTICS'

WHAT IT ENCODES

FSGS is the most common primary glomerular disease causing end-stage kidney disease (ESKD) in the US and accounts for roughly one-third of adult nephrotic syndrome cases (the single most common cause of nephrotic syndrome in Black adults). 'Focal' = some glomeruli affected; 'segmental' = only part of each affected tuft is sclerosed — so sampling matters and a biopsy can miss it. Light microscopy shows segmental capillary collapse with hyalinosis; IF is usually negative or nonspecific (IgM/C3 trapping).

BOARD TRAP

WRONG ANSWER: 'membranous nephropathy is the most common primary cause of nephrotic syndrome / ESKD in adults.' Discriminator: membranous is the classic cause in OLDER white adults, but FSGS is overall the most common primary glomerular disease progressing to ESKD in the US and the leading nephrotic pattern in Black adults. Match the demographic: Black adult with nephrotic syndrome → FSGS, not membranous.

2. African ancestry 50%

WHAT YOU SEE

Sign 'AFRICAN ANCESTRY 50%'

WHAT IT ENCODES

Strong predilection for patients of African ancestry, driven largely by APOL1 high-risk variants (G1/G2) on chromosome 22; two risk alleles markedly raise lifetime risk of FSGS, HIVAN, and hypertensive nephrosclerosis. This explains the higher incidence and faster progression of FSGS in Black patients. The same APOL1 biology underlies the high-risk donor-kidney concept in transplant.

BOARD TRAP

WRONG ANSWER: 'FSGS is most common in elderly white patients.' Discriminator: elderly white = think membranous; FSGS skews toward younger and Black patients via APOL1 risk variants. Picking 'elderly white' confuses FSGS with membranous nephropathy.

3. Histologic variants

WHAT YOU SEE

Framed gallery: NOS, TIP LESION, CELLULAR, PERIHILAR, COLLAPSING

WHAT IT ENCODES

The Columbia classification recognizes five morphologic variants: NOS (not otherwise specified, most common), perihilar (typical of secondary/adaptive forms), cellular, tip lesion (best prognosis, often steroid-responsive), and collapsing (worst prognosis, seen in HIVAN and APOL1/drug-related disease). Variant predicts prognosis and treatment response, so the pathology read matters clinically.

BOARD TRAP

WRONG ANSWER: 'FSGS is a single uniform histologic pattern.' Discriminator: FSGS is a heterogeneous family of five Columbia variants with very different prognoses (tip best, collapsing worst). A stem that treats all FSGS identically is wrong — the variant (and primary vs secondary etiology) drives both prognosis and whether to immunosuppress.

4. Collapsing = worst

WHAT YOU SEE

'COLLAPSING' frame with skull

WHAT IT ENCODES

The collapsing variant shows global/segmental capillary tuft collapse with overlying podocyte hyperplasia and is the most aggressive form, with the worst prognosis and rapid progression to ESKD. It is the hallmark of HIVAN and is also seen with APOL1 risk variants, interferon therapy, pamidronate, and parvovirus B19.

BOARD TRAP

WRONG ANSWER: 'the tip lesion variant carries the worst prognosis.' Discriminator: it's the reverse — the TIP lesion has the BEST prognosis (often steroid-responsive), while the COLLAPSING variant has the worst. If a stem describes an HIV-positive patient with rapid renal decline, expect collapsing FSGS, not tip lesion.

5. Foot process effacement

WHAT YOU SEE

Bishop chess piece / podocyte detail near board

WHAT IT ENCODES

FSGS is fundamentally a PODOCYTOPATHY: electron microscopy shows diffuse podocyte foot process effacement (extensive in primary, more focal/partial in secondary adaptive forms). Like minimal change disease it is a podocyte injury disease, but FSGS adds segmental scarring on LM — many consider MCD and primary FSGS a spectrum.

BOARD TRAP

WRONG ANSWER: 'subendothelial immune-complex deposits drive FSGS' (i.e., picking an immune-complex/MPGN or lupus mechanism). Discriminator: FSGS is a non-immune-complex podocytopathy — EM shows foot process effacement, and IF is negative or only shows nonspecific IgM/C3 trapping in sclerotic segments. Subendothelial deposits point to MPGN or class IV lupus nephritis, not FSGS.

6. Primary = permeability factor

WHAT YOU SEE

Angry pink 'PERMEABILITY FACTOR' creature, 'PRIMARY' arrow

WHAT IT ENCODES

Primary (idiopathic) FSGS is thought to be driven by a circulating permeability factor (candidates include suPAR and anti-nephrin/CD40 antibodies) that injures podocytes diffusely. It presents abruptly with the FULL nephrotic syndrome — heavy proteinuria (often >3.5 g/day, can be massive), hypoalbuminemia, edema, and diffuse foot process effacement on EM. This abrupt nephrotic onset with diffuse effacement is what marks it as primary and immunosuppression-responsive.

BOARD TRAP

WRONG ANSWER: 'primary FSGS is caused by glomerular hyperfiltration/adaptive injury.' Discriminator: hyperfiltration/reduced nephron mass is the mechanism of SECONDARY FSGS. Primary FSGS is the circulating-permeability-factor podocytopathy with abrupt full-blown nephrotic syndrome and diffuse foot process effacement — and only primary FSGS warrants immunosuppression.

7. Secondary: adaptive/hyperfiltration

WHAT YOU SEE

'SECONDARY' arrow, obesity/reduced nephron cues

WHAT IT ENCODES

Secondary FSGS results from adaptive glomerular hyperfiltration or reduced nephron mass: obesity, vesicoureteral reflux, prior nephrectomy/oligonephronia, sickle cell disease, chronic hypertension, and healed prior GN. It typically presents with slowly rising, often SUB-nephrotic proteinuria, preserved serum albumin, and only segmental (not diffuse) foot process effacement on EM. Drug- and virus-associated forms (heroin, pamidronate, anabolic steroids, interferon) also fall under non-primary FSGS.

BOARD TRAP

WRONG ANSWER: 'secondary FSGS always presents with full nephrotic syndrome (heavy edema + hypoalbuminemia).' Discriminator: secondary/adaptive FSGS usually causes sub-nephrotic proteinuria with NORMAL serum albumin and minimal edema because protein loss develops gradually — full nephrotic syndrome with hypoalbuminemia favors PRIMARY FSGS. Mislabeling secondary as primary leads to inappropriate immunosuppression.

8. Genetic FSGS

WHAT YOU SEE

DNA helix 'GENETIC' arrow; NPHS1/NPHS2/podocin genes panel

WHAT IT ENCODES

Genetic/familial FSGS arises from mutations in podocyte structural/slit-diaphragm genes: NPHS1 (nephrin, congenital nephrotic syndrome of the Finnish type), NPHS2 (podocin), ACTN4 (alpha-actinin-4), TRPC6, WT1, INF2. Suspect it in congenital/childhood onset, positive family history, syndromic features, or steroid resistance — genetic testing changes management and avoids futile immunosuppression.

BOARD TRAP

WRONG ANSWER: 'genetic FSGS responds well to corticosteroids.' Discriminator: monogenic podocyte-gene FSGS is characteristically STEROID-RESISTANT (the defect is structural, not a circulating factor), and immunosuppression is generally avoided. Steroid resistance plus young age or family history should trigger genetic testing rather than escalating immunosuppression.

9. Steroids 1st line (primary)

WHAT YOU SEE

Chess king 'STEROIDS 1ST LINE / 6 MONTHS / APOL1'

WHAT IT ENCODES

For PRIMARY FSGS, prolonged high-dose corticosteroids are first line — e.g., prednisone ~1 mg/kg/day (max ~80 mg) often continued for up to 4–6 months because remission is slow; steroid-resistant cases get calcineurin inhibitors (cyclosporine or tacrolimus) as the main second-line, with adjunctive RAAS blockade throughout.

BOARD TRAP

WRONG ANSWER: 'immunosuppression is never indicated in FSGS' OR 'immunosuppress every FSGS patient.' Discriminator: the answer is conditional — immunosuppress PRIMARY FSGS (prolonged steroids first line, CNi for steroid resistance), but do NOT immunosuppress secondary or genetic FSGS. The board point is matching therapy to etiology, not a blanket yes/no.

10. Don't immunosuppress secondary

WHAT YOU SEE

Red STOP sign 'NO IMMUNOSUPP / RAAS'

WHAT IT ENCODES

Secondary (adaptive/maladaptive) FSGS should NOT be treated with steroids/immunosuppression — there is no circulating immune driver. Management is conservative/antiproteinuric: maximal RAAS blockade (ACEi or ARB), blood-pressure and sodium control, weight loss for obesity-related disease, and stopping the offending drug.

BOARD TRAP

WRONG ANSWER: 'immunosuppress all FSGS regardless of cause.' Discriminator: in secondary FSGS the podocyte injury is from hyperfiltration, so immunosuppression adds toxicity without benefit — RAAS blockade and treating the underlying cause are correct. Reflexively giving steroids to an obese patient with sub-nephrotic FSGS is the classic trap.

11. RAAS + SGLT2i all types

WHAT YOU SEE

Banner 'RAAS + SGLT2i — ALL TYPES'

WHAT IT ENCODES

Foundational antiproteinuric therapy applies to ALL FSGS regardless of cause: ACEi or ARB to lower intraglomerular pressure and proteinuria, now paired with an SGLT2 inhibitor (e.g., dapagliflozin), which reduces proteinuria and slows CKD progression in proteinuric kidney disease. Add BP control and dietary sodium restriction; these are background therapy on top of (or instead of) immunosuppression depending on etiology.

BOARD TRAP

WRONG ANSWER: 'RAAS blockade is contraindicated in heavy proteinuria.' Discriminator: the opposite is true — ACEi/ARB are first-line antiproteinuric agents and reduce progression in proteinuric glomerular disease; they are cornerstone therapy in FSGS (watch potassium/creatinine, but a modest creatinine rise is acceptable). SGLT2 inhibitors are an additive, not contraindicated, therapy.

12. Secondary → treat underlying

WHAT YOU SEE

'SECONDARY: TREAT UNDERLYING' tree banner

WHAT IT ENCODES

For secondary FSGS, treat the root cause: weight loss/bariatric measures for obesity-related FSGS, correct reflux, discontinue the offending drug (pamidronate, anabolic steroids, interferon, heroin), control hypertension, and optimize sickle cell disease — alongside RAAS/SGLT2i. Etiology-directed therapy, not immunosuppression, is the answer.

BOARD TRAP

WRONG ANSWER: 'treat secondary FSGS with high-dose corticosteroids.' Discriminator: secondary FSGS has no immune target — steroids are ineffective and harmful; you remove the driver (lose weight, stop the drug) plus give RAAS blockade. Only PRIMARY FSGS gets steroids.

13. Recurs after transplant

WHAT YOU SEE

'TRANSPLANT GATE' / washing machine plasma exchange

WHAT IT ENCODES

Primary FSGS recurs in the renal allograft in roughly 30–40% of cases (higher with prior recurrence, rapid native progression, younger age), often within days to weeks because of the circulating permeability factor. Treatment of recurrence is plasmapheresis/plasma exchange (to remove the factor) ± rituximab and intensified immunosuppression — the 'washing machine.'

BOARD TRAP

WRONG ANSWER: 'FSGS never recurs after kidney transplant' (or 'genetic FSGS recurs at high rates'). Discriminator: PRIMARY FSGS recurs frequently post-transplant (circulating factor), treated with plasma exchange; GENETIC FSGS rarely recurs because there's no circulating factor — the recipient's transplanted podocytes have normal genes. Match recurrence risk to the primary-vs-genetic mechanism.

14. ~50% ESKD in 10 yrs

WHAT YOU SEE

Ghost circle '–50% ESKD in 10 YRS'

WHAT IT ENCODES

Prognosis hinges on remission: untreated or steroid-resistant primary FSGS with persistent nephrotic-range proteinuria progresses to ESKD in roughly 50% of patients within ~10 years, whereas achieving complete/partial remission dramatically improves renal survival. Degree of proteinuria and remission status are the key prognostic determinants.

BOARD TRAP

WRONG ANSWER: 'FSGS rarely progresses to kidney failure / has a benign course.' Discriminator: FSGS is a leading cause of ESKD precisely because refractory nephrotic FSGS commonly progresses (~50% to ESKD in a decade if proteinuria persists). Calling it benign ignores its status as the #1 primary glomerular cause of ESKD in the US.